

FA-BENCH: A BENCHMARK FOR PHONE- AND WORD-LEVEL TIMESTAMP ACCURACY IN FORCED ALIGNMENT AND ASR ON CLEAN AND NOISY SPEECH

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ABSTRACT

Forced alignment estimates the timestamps of each word, phone or character in speech given its transcript. Published comparisons normalize transcripts, split the data and match boundaries differently, so their numbers cannot be read together. We present **FA-Bench**, an open framework that fixes those choices once and releases the code, splits, phone mapping, text normalization and scoring script, with results published periodically. Track 1 gives every aligner the reference transcript and Track 2 gives it a recognizer’s output, on the same audio, clean and degraded four ways, with 21 open models and 9 commercial APIs under a unified protocol. We score every boundary of an utterance and check the two labels beside it, so a word the recognizer missed or invented is charged. Using a tolerance-based F_1 as our primary metric eliminates the 9% to 14% score inflation that standard MAE causes on recognition-dependent systems in conversational speech. We then group boundaries by position and how many adjacent words were recognized correctly, which shows where a system lost the score. We discovered systematic bias in how current systems time words, with Whisper about 150ms early and several commercial ASR APIs over 50ms late. Code and results are at <https://github.com/olewave/fa-bench>.

Index Terms— word timestamps, forced alignment, benchmark, noise robustness, speech corpora

1. INTRODUCTION

Given a transcript, forced alignment estimates where each of its words, phones or characters starts and ends in the speech audio. HMM-GMM systems such as P2FA [1], Prosodylab-Aligner [2] and MFA [3] are still in use, and neural frame classifiers [4, 5], CTC aligners over self-supervised encoders [6, 7], attention models [8] and end-to-end differentiable aligners [9] have followed. Timestamped ASR systems [10, 11, 12] produce times for the words they decode, and have been compared only within the Whisper family [12, 13].

Their published numbers cannot be read against each other, because every evaluation normalizes transcripts, splits

the data and matches boundaries its own way, and most use clean speech. Some treat another aligner’s boundaries as ground truth [10, 12, 13, 14, 15, 16, 17]. Buckeye ships no official split, so every paper picks its own [5, 8, 9], and one paper’s held-out speaker is another’s training speaker.

We build FA-Bench for word and phone forced alignment, and also for evaluating the timestamps output from ASR systems and forced alignment using ASR results as reference words. In practice the words to align usually come from a recognizer. We make four contributions.

- **An open framework for researchers to exactly reproduce results and for developers to objectively compare systems.** For researchers, we release the code, splits, phone mapping, text normalization and scoring script, publish benchmark results periodically, and a new system is one wrapper away. For developers choosing a service or model from many that come with no evaluation, the published records/¹ are where to look, and the harness runs on their own proprietary data.
- **Scoring every boundary and examining the two labels adjacent to it.** MFA’s evaluation [18] assumes a gold transcript and never checks the labels beside a boundary, so once the words come from ASR, a boundary between two misread units still counts. It also skips the first and last boundary of each utterance, 18–22% of the word boundaries here. Rouso et al. [19] and AssemblyAI [15] scored matched words only, so a recognizer pays nothing for a word it drops. Yeh et al. [13] score every word but check only its end, which misses a late start.
- **Grouping boundaries into five categories for analyzing where an ASR’s timestamps lost the score.** A single F_1 or MAE cannot tell a system that misreads words from one that times correct words badly. It also averages utterance edges with interior boundaries as if they were one problem. An edge is found against silence, and an interior boundary sits between two words. So we class each boundary by where it sits and how many adjacent words match the reference.
- **Discovering significant systematic bias in the word start and end times of current open systems and APIs.** Fig. 1 shows it and Sec. 3 discusses it.

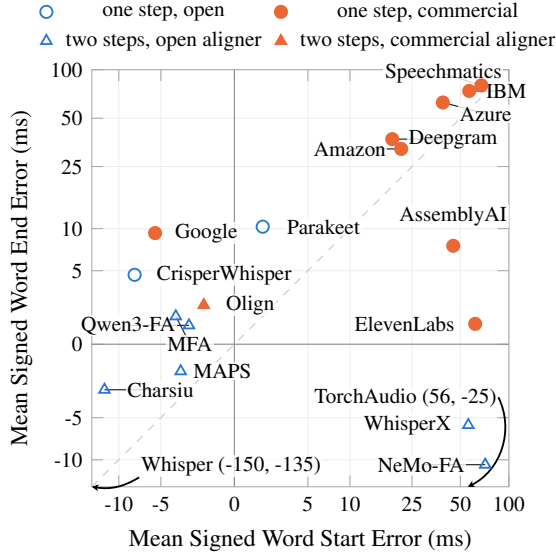


Fig. 1. Mean signed word-boundary error on TIMIT test, hypothesis minus reference, over the words the label alignment matched. Track 2 only. Every two-step point aligns Qwen3-ASR’s transcript, so the recognizer is fixed and only the aligner varies. Both axes are warped by $\text{asinh}(v/4)$. The distance from the dashed line $y = x$ is the mean error in word duration.

2. THE BENCHMARK

Every cell of every system is published in `records`¹ with its metrics.

2.1. Tasks

FA-Bench evaluates a word tier and a phone tier under two tracks. Track 1 measures timing alone and Track 2 timing plus whatever the recognizer got wrong, so the two blocks answer different questions.

	Track 1	Track 2	
	reference transcript	one step ASR with timestamps	two steps align after ASR
Word tier	✓	✓	✓
Phone tier	✓	×	✓ ¹

Open and closed systems. The versions and settings of the 21 open models and the 9 APIs are documented in `records`¹. The company of the first group of authors offers Oligin API [20], which is not trained with dev or test splits of TIMIT and Buckeye.

2.2. Data

Corpora. We use TIMIT [21] for read speech and Buckeye [22] for conversational speech, both with word and phone

boundaries marked by hand. Phone tiers were annotated from the audio rather than expanded from pronunciation dictionaries. In Buckeye, the canonical and realized pronunciations differ for 58.7% of word tokens. A dictionary-based aligner therefore proposes sounds the talker never made.

Splits. All evaluation speakers are held out. TIMIT uses the corpus core test set and Kaldi’s development split. Buckeye has no official split, so we define one while preserving its sex×age structure. Each of the four younger/older × female/male cells contributes six speakers to training and two each to development and test.

Transcript normalization. TIMIT .WRD transcripts are lower-cased without further modification. Buckeye sometimes hides a spoken word inside an event tag such as <LAUGH-here>. We take the word back when the phone tier under the tag holds real speech, keeping the annotated times, and drop the tag when it does not. Because aligners may renormalize their input to match their lexicons, their outputs are mapped back to the input tokens whenever the letters spell that token back, ignoring case and punctuation.

Degradations. We follow the Kaldi [23] VoxCeleb recipe and its default parameters, which add MUSAN [24] noise, music and babble, all of them recorded, together with simulated room impulse responses [25].

2.3. Scoring

Shared phone inventory. Each system aligns in its own phone set, and the two corpora do not share one either. We map its phones and the reference phones into TIMIT-39 [26] afterwards, the conventional reduction for these corpora and the one Kaldi ships [23]. The map exists only so the rows can be read against each other. It renames a unit and never moves a boundary, and the only label we drop is the glottal stop, 1.6% of TIMIT phones and none of Buckeye’s. The source alphabets and the mappings are in `records`¹. What costs these systems is the segmentation convention, e.g. `bcl` before `b`. Although we reported numbers with the closure mapped as silence, FA-Bench as a tool still offers users the option of merging the closure into its burst.

Metrics. In FA-Bench, a *matched boundary* is defined as a detected boundary whose adjacent labels match the ground truth labels. The label can be word or phone. Note that for Track 1, where the reference words are available, every word boundary is a matched boundary. For Track 2, the matched boundaries are obtained by aligning recognized and reference words first. Then, the *boundary mean absolute error (MAE)* is calculated by averaging the absolute difference in ms between their times and the ground truth. In Track 2, due to recognition errors, not all ground truth boundaries are used in the MAE calculation. Qwen3-ASR makes 13.5% word errors on noisy Buckeye, so its MAE is computed on 83% of the reference boundaries.

A recognizer can drop the words it finds hard and gain

¹Full records are at <https://github.com/olewave/fa-bench/tree/main/records>

from doing so, since MAE then reports only the easy words it kept. Scored only at the words a Track 2 recognizer got right, Olign’s Buckeye MAE drops by 9% to 14%, while its TIMIT MAE does not move. Pitch tracking has the same trade-off, where a low gross pitch error can come with a high voicing error, so F0 frame error counts both [27]. To count every reference boundary we use *boundary* F_1 , whose recall takes every reference boundary and whose precision takes every hypothesized one. A boundary is a hit only when it is a matched boundary and its time falls within the tolerance of the ground truth. The two utterance edges are boundaries as well, against silence, and silence matches, so an edge answers to its one word. A word the system misread therefore costs it the two boundaries beside that word. An utterance the system returns nothing for costs it every boundary in that utterance, and every word and phone in it counts as deleted. Tables 1 and 2 report 20ms. The tolerance is swept from 10 to 100 ms in *records*/¹, with over-segmentation (OS) and R-value [28].

Grouped boundaries. We wonder how recognition errors affect the accuracy of forced alignment, which includes what a single mismatched label costs against two matched labels, whether an utterance edge is harder than an interior boundary, whether finding where speech begins differs from finding where it ends, and what is left once both labels are wrong. So we group every boundary by where it sits and whether its adjacent labels match, into 5 categories (Table 3).

3. RESULTS

Word and phone tier results are shown in Tab. 1 and 2. The results of the dev sets and more details are in *records*/¹.

Clean-speech MAE rankings do not predict robustness. MAPS [5] is third of the fourteen Track 1 aligners on clean TIMIT word MAE and last under degradation, and four of the fourteen move three or more places on TIMIT and seven on Buckeye. The F_1 ranking barely moves, because a gross error inflates MAE but is only one miss for F_1 .

Phone error rate measures the lexicon. All Track 1 systems in Table 2 are fed the reference word sequence, except FALCON, which has no English G2P [9] and is fed the reference phones instead, and thus leads the phone F_1 on TIMIT. For the others, the clean-audio PER of 26 to 36% measures a dictionary against a hand transcription. Degrading the audio moves the PER column by 0.3 points or less for six of the eight systems on TIMIT and five on Buckeye while their phone MAE moves by up to 82 ms. Since both phones beside a boundary must match, the same mismatch caps the phone F_1 , and only 43 to 54% of the TIMIT phone boundaries have both neighbours matched.

Commercial APIs can suffer from conservativeness. Under noisy conditions, a conservative aligner could loosen utterance boundaries into silence to prevent possibly cutting the edge words, which can cost its edge boundary precision. Olign places interior word boundaries better than MFA,

Table 1. Word-tier boundary error on the test split of each corpus, on clean audio and under degradation, *noisy* being the mean of the four. Every number here comes from our own runs. MAE is over word boundaries in ms. F_1 is over every word boundary, the two utterance edges included. A boundary is a hit only when the words on each side of it match the reference and the time falls within the tolerance of the reference boundary. The superscript on F_1 is that tolerance in ms. The step size of a word boundary of each system is shown in Table 3. Open means the weights are published, and NeMo-FA, WhisperX, TorchAudio, BFA, MMS-FA, UnitY2, Charsiu, MFA and MAPS also publish the code that trained them. The NeuFA model is trained with its open-source code on GitHub and is not released. W is Whisper large-v3, P and Parakeet are Parakeet-TDT, Q and Qwen3 are Qwen3-ASR, and G is Google Chirp 2. Whisper-ts is Whisper-timestamped and NeMo-FA uses its Conformer checkpoint. The APIs are Speechmatics Enhanced, Deepgram Nova-3, IBM Large US English, Azure en-US, AssemblyAI Universal 3.5 Pro, Amazon Transcribe en-US, ElevenLabs Scribe v2, Google Chirp 2 and Olign 0.9. MFA is Montreal Forced Aligner 3.4. MAPS and NeuFA trained on 7 of 8 speakers in both Buckeye splits.

Family	System	TIMIT test				Buckeye test				
		clean		noisy		clean		noisy		
Track 1, given the reference transcript										
		MAE	F_1^{20}	F_1^{50}	MAE	F_1^{20}	F_1^{50}	MAE	F_1^{20}	F_1^{50}
Open	NeMo-FA [14]	47.4	.17	.41	47.1	.18	.43	62.8	.21	.49
	stable-ts [29]	96.8	.18	.39	94.5	.19	.41	71.2	.29	.55
	WhisperX [10]	37.2	.18	.43	39.5	.17	.42	41.7	.16	.49
	TorchAudio [30]	39.8	.21	.47	42.4	.19	.45	42.4	.18	.53
	BFA [7]	51.0	.27	.62	56.7	.24	.57	57.2	.30	.60
	MMS-FA [31]	31.1	.28	.64	31.5	.26	.62	37.1	.23	.64
	UnitY2 [32]	52.2	.34	.69	53.8	.35	.68	40.6	.48	.79
	Qwen3-FA [16, 17]	32.2	.40	.82	38.5	.38	.78	33.8	.43	.84
	CrisperW [12]	34.1	.44	.79	36.9	.43	.77	43.1	.49	.78
	NeuFA [8]	56.9	.52	.70	88.6	.45	.64	33.9	.70	.86
	Charsiu [4]	28.1	.54	.86	40.2	.49	.79	28.1	.61	.87
	MFA [3]	21.8	.64	.92	30.2	.62	.86	21.3	.68	.93
	MAPS [5]	25.5	.71	.89	110.0	.40	.59	37.9	.62	.84
	API	Olign [20]	18.2	.78	.91	33.0	.67	.82	22.5	.74
Track 2, one step. The ASR times its own words										
		MAE	F_1^{20}	WER	MAE	F_1^{20}	WER	MAE	F_1^{20}	WER
Open	Whisper [11]	148.0	.10	2.9	151.8	.10	3.2	122.7	.14	13.7
	Whisper-ts [33]	161.6	.11	2.5	163.9	.11	3.4	133.3	.17	13.4
	TorchAudio	37.8	.17	10.8	38.2	.12	25.7	37.8	.08	27.8
	Parakeet [34]	79.3	.18	2.4	77.1	.18	3.5	80.7	.19	11.0
	CrisperW	33.5	.42	3.6	36.2	.41	4.3	37.6	.43	11.5
	Azure [35]	64.4	.14	3.4	67.4	.14	4.6	84.7	.14	11.5
API	IBM [36]	72.4	.15	5.1	75.9	.14	9.2	62.0	.18	12.4
	Speechmatics [37]	77.1	.15	2.5	81.5	.14	3.4	74.2	.14	11.4
	ElevenLabs [38]	32.1	.17	2.0	32.6	.16	3.4	34.8	.11	10.6
	Deepgram [39]	67.6	.21	2.4	70.2	.20	4.4	97.8	.20	10.8
	AssemblyAI [40]	47.9	.24	3.7	53.9	.22	4.2	56.4	.20	12.2
	Amazon [41]	55.7	.25	2.7	56.6	.26	2.9	52.8	.29	10.3
API	Google [42]	32.7	.44	4.2	32.6	.45	6.7	26.9	.44	13.1
two steps. → denotes ASR then aligner										
Open	W→WhisperX	35.1	.18	2.9	38.4	.16	6.6	46.2	.13	14.2
	Q→Qwen3-FA	32.6	.39	2.0	38.3	.37	2.4	30.6	.39	10.2
	Q→MFA	21.8	.63	2.0	29.6	.61	2.8	22.7	.60	10.4
	Q→Olign	18.1	.76	2.0	34.2	.61	2.4	20.6	.67	10.2
API	G→Olign	18.1	.73	4.2	33.7	.57	6.7	20.2	.63	13.1
	P→Olign	18.2	.75	2.4	33.8	.60	3.5	20.1	.67	10.7

9.3 ms against 14.3 ms clean and 15.0 ms against 19.7 ms degraded, and overshoots into the silence at a word end, 6 ms clean and 25 ms degraded where MFA overshoots 14 ms and 8 ms. That flips its noisy TIMIT MAE to 33.0 ms against

Table 2. Phone-tier boundary error on the test split of each corpus, on clean audio and under degradation, *noisy* being the mean of the four, laid out as Table 1. Every Track 1 aligner is given the reference words and derives its own phones, and PER is the phone error rate of what it derived against the gold phones, per split and condition. FALCON, daggered, is handed the gold phones, and overlaps our Buckeye split in its training data. We use the FALCON released model and did not retrain it. No one-step ASR in Table 1 emits phone timestamps, so Track 2 has only the two-step rows whose aligner does, and their PER carries the recognizer’s word errors too.

Family	System	TIMIT test						Buckeye test					
		clean			noisy			clean			noisy		
		Track 1, reference words, except FALCON[†]											
Open		MAE	F_1^{20}	PER	MAE	F_1^{20}	PER	MAE	F_1^{20}	PER	MAE	F_1^{20}	PER
	BFA [7]	45.1	.18	36.2	48.9	.18	36.2	50.2	.25	32.1	54.8	.25	32.2
	TorchAudio [30]	33.8	.21	35.9	33.3	.21	35.9	34.5	.27	28.5	41.3	.26	28.5
	NeuFA [8]	20.5	.33	36.0	56.1	.28	39.1	22.6	.46	28.0	46.5	.40	30.7
	Charsiu [4]	19.9	.36	30.3	28.3	.35	31.1	21.1	.42	31.7	50.4	.36	34.8
	MFA [3]	11.6	.39	34.8	17.0	.37	35.0	15.3	.52	26.1	26.1	.46	29.5
	MAPS [5]	15.3	.45	30.7	89.6	.32	30.7	22.9	.44	34.9	104.9	.31	34.9
	FALCON [†] [9]	22.3	.67	0.0	67.8	.52	0.0	40.6	.58	0.7	99.5	.46	0.7
API	Olign [20]	8.4	.41	35.4	13.1	.39	35.5	12.7	.52	28.2	24.2	.48	28.5
Track 2, two steps. → denotes ASR then aligner													
Open		MAE	F_1^{20}	PER	MAE	F_1^{20}	PER	MAE	F_1^{20}	PER	MAE	F_1^{20}	PER
	Q→MFA	11.6	.39	34.8	16.6	.37	35.2	17.6	.50	28.2	27.2	.45	32.4
API	Q→Olign	7.5	.41	35.4	12.8	.39	35.6	12.4	.52	29.2	22.6	.46	31.0
	G→Olign	8.3	.41	36.2	13.2	.38	36.7	12.4	.51	30.3	22.0	.44	33.4
	P→Olign	7.5	.41	35.4	12.7	.39	35.9	12.3	.52	29.1	21.8	.46	31.8

Table 3. The F_1 s at 20 ms of different types of word-tier boundaries of Track 2 on Buckeye test, on clean audio and under degradation, *noisy* being the mean of the four. A boundary inside (*I*) an utterance has two adjacent words. We use $M2_I$ to denote both words are matched and $M1_I$ to denote one label is matched, where *M* denotes matched. $M1_B$ and $M1_E$ denote the beginning (*B*) and the ending (*E*) boundary of an utterance, each with only one adjacent word. $M0$ denotes the remaining boundaries. The first row is the share of reference boundaries in each category, averaged over the rows.

	Grid (ms)	Buckeye test clean						Buckeye test noisy					
		$M2_I$	$M1_I$	$M1_B$	$M1_E$	$M0$	$M2_I$	$M1_I$	$M1_B$	$M1_E$	$M0$		
		% of boundaries	69.2	10.4	7.2	7.6	5.5	64.4	12.2	6.7	6.9	9.7	
Open	Whisper-ts	20	.17	.08	.58	.03	.14	.17	.09	.57	.02	.14	
	Whisper	20	.12	.06	.57	.05	.13	.12	.07	.57	.04	.14	
	Parakeet	80	.23	.15	.22	.04	.08	.23	.16	.23	.03	.09	
	W→WhisperX	20	.13	.07	.10	.28	.06	.13	.08	.08	.25	.08	
	TorchAudio	20	.13	.13	.14	.09	.15	.12	.11	.11	.09	.13	
	CrisperW	10	.52	.34	.56	.21	.20	.47	.31	.55	.22	.22	
	Q→Qwen3-FA	80	.43	.30	.55	.35	.21	.40	.26	.52	.27	.17	
	Q→MFA	10	.73	.47	.57	.33	.22	.64	.36	.56	.22	.18	
API	Deepgram	80	.21	.12	.48	.14	.16	.21	.13	.48	.14	.14	
	Speechmatics	40	.18	.10	.16	.01	.06	.17	.09	.16	.01	.06	
	Azure	10	.17	.12	.09	.11	.06	.16	.10	.08	.11	.06	
	IBM	20	.24	.16	.14	.01	.07	.22	.15	.12	.02	.05	
	Amazon	10	.33	.29	.62	.02	.20	.32	.28	.62	.02	.18	
	AssemblyAI	1	.23	.14	.39	.09	.17	.22	.12	.34	.08	.14	
	ElevenLabs	20	.12	.10	.13	.09	.09	.13	.10	.14	.08	.10	
	Google	40	.58	.44	.13	.43	.18	.58	.42	.17	.39	.17	
	Q→Olign	10	.80	.50	.58	.50	.27	.70	.41	.46	.31	.19	
	G→Olign	10	.80	.53	.59	.50	.31	.71	.44	.47	.31	.21	
	P→Olign	10	.80	.57	.59	.50	.29	.71	.47	.46	.31	.20	

30.2 ms. Averaged over clean and degraded, Olign overshoots by 15 ms and Google Chirp 2 by 56 ms.

Re-aligning ASR output improves over the bundled aligner. Track 2 hands the recognizer’s words to the aligner,

and its errors come along but its timestamps need not. Google Chirp 2 followed by Olign [20] improved the F_1 at 20 ms by 40% on average over the four test cells.

ASR accuracy affects FA accuracy. Recognition errors directly lower the alignment score by leaving boundaries with only one matched label. These $M1_I$ boundaries from Table 3 never count as hits in the overall F_1 of Tables 1 and 2. When average Track 2 WER on Buckeye worsens from 12.6% clean to 17.7% degraded, this unscorable $M1_I$ group expands from 10.4% to 12.2%. Aligners also time boundaries much better when they recognize both adjacent words. The mean F_1 across Track 2 systems on clean Buckeye is 0.36 for $M2_I$ boundaries and only 0.25 for $M1_I$. A stronger recognizer fixes both problems by shifting boundaries into $M2_I$ where they count and are timed more accurately.

Utterance end timestamps are more difficult to estimate than beginning and internal ones. As shown in Table 3, the boundary detection accuracy is related to the location of the boundary in an utterance. Averaged over the Track 2 systems on Buckeye test, the F_1 of utterance end boundaries is 0.18 and 0.17 lower than that of beginning and internal boundaries respectively.

Word start and end times carry large systematic bias. As shown in Fig. 1, a system’s mean signed error is rarely near zero. Whisper is about 150 ms early and seven of the nineteen plotted are more than 50 ms out at the word start. Absolute error cannot see this. Whisper sits on the dashed diagonal, about as early at both ends, so its word durations are right to within 16 ms and a duration check would pass timestamps 150 ms early. Seven of the eight commercial ASR APIs start their words late, three of them by more than 50 ms.

A coarse timestamp grid caps detection. Some systems report times only in 80 ms steps, the Grid column of Table 3. Only half of the reference boundaries fall within 20 ms of such a step, so a forced aligner on that grid is statistically capped near an F_1 of 0.5 at 20 ms. Qwen3-FA reports on 80 ms and its 0.40 in Table 1 sits under that cap.

4. CONCLUSIONS

FA-Bench is a common evaluation setting for forced alignment and for the timestamps an ASR system emits. It fixes transcript normalization, held-out splits and the phone inventory, and releases the scoring code. By running FA-Bench on human-annotated datasets, we show that word error rate alone does not determine timestamp accuracy, that a clean-speech MAE ranking does not survive degradation, and that word boundaries carry offsets that absolute error never sees. Every boundary is scored, utterance edges included, and a boundary counts only when both labels beside it match the reference. Grouping boundaries by where they sit and by how many adjacent labels match turns one number into a statement about where a system lost it. Future releases will add more languages and systems, and more options for users.

5. REFERENCES

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